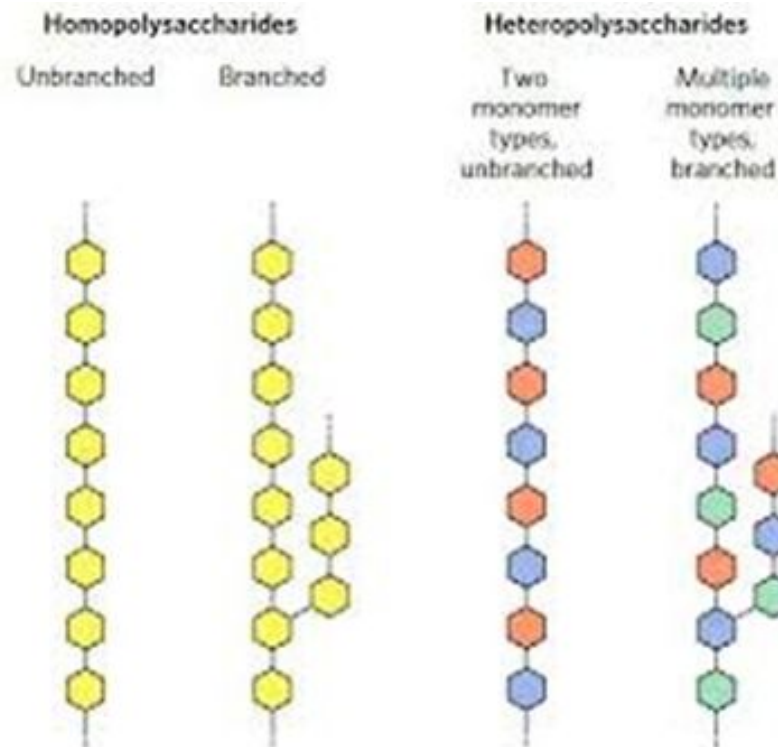


HETEROPOLYSACCHARIDES or HETEROGLYCANS



Dr. Humaira Aman Ali

Polysaccharides

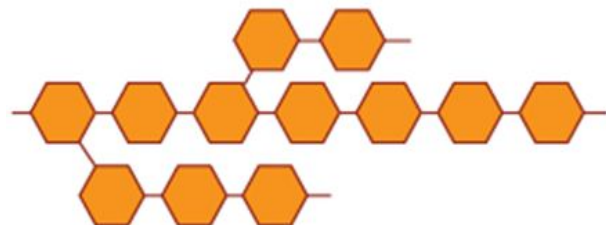
Homopolysaccharides

one type of repeating unit

Homopolysaccharides



Unbranched



Branched

Heteropolysaccharides

more than one type of repeating units

Heteropolysaccharides



Unbranched



Branched

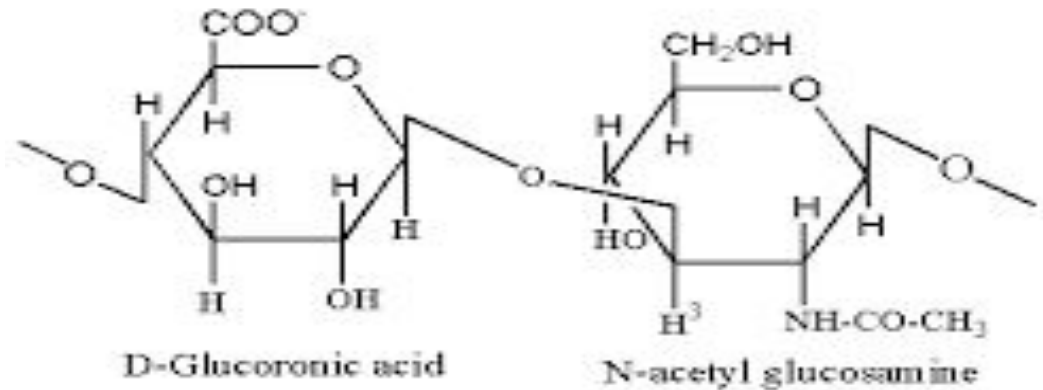
Heteropolysaccharides

- Polymers made from more than one kind of monosaccharides or monosaccharide derivatives.

- Eg : Glycosoaminoglycans (GAGs)

Agar

Agarose



Hyaluronic acid

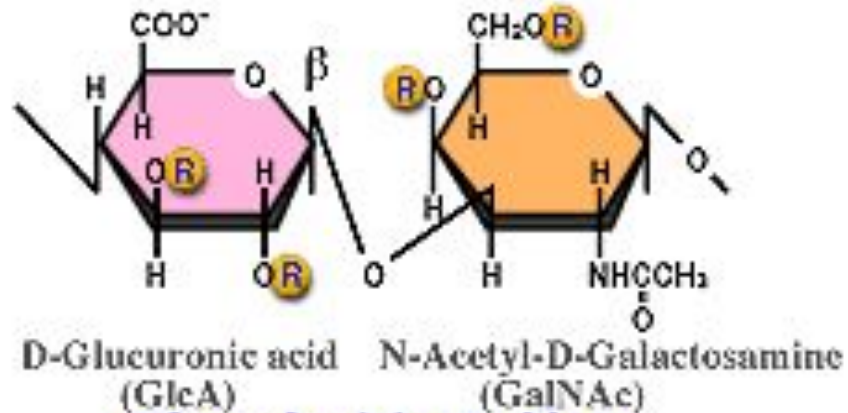
GLYCOSAMINOGLYCANS OR MUCOPOLYSACCHARIDES

- ✓ Glycosaminoglycans (GAGs) are primarily known as **mucopolysaccharides** because isolated first time from mucin.
- ✓ As highly polar, negatively charged polysaccharides, they are also frequently referred to as **acid mucopolysaccharides** due to their acidic nature.
- ✓ GAGs are the most important group of heteroglycan in humans.

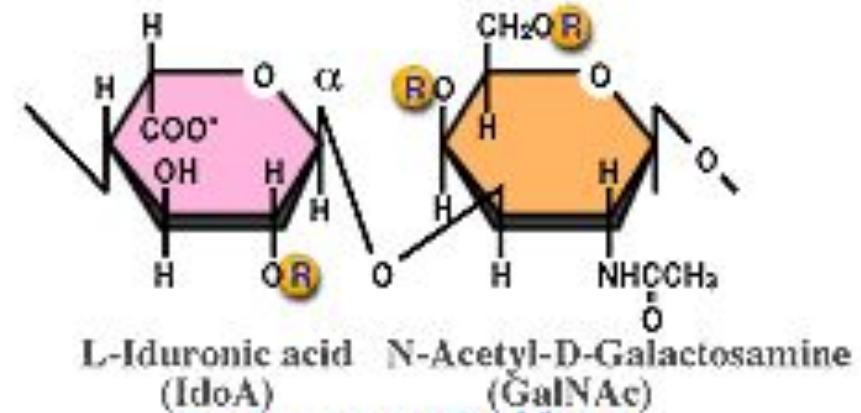
cont..

- ✓ Glycosaminoglycans are **long, linear** polysaccharides with **repeating disaccharide** units.
- ✓ Unbranched heteropolysaccharide, made of repeating disaccharide units containing uronic acid & amino sugars.
- ✓ Amino sugar – Glucosamine or Galactosamine
 - ✓ (Present in there acetylated form)
- ✓ Uronic acid – D-Glucuronic acid or L-Iduronic acid

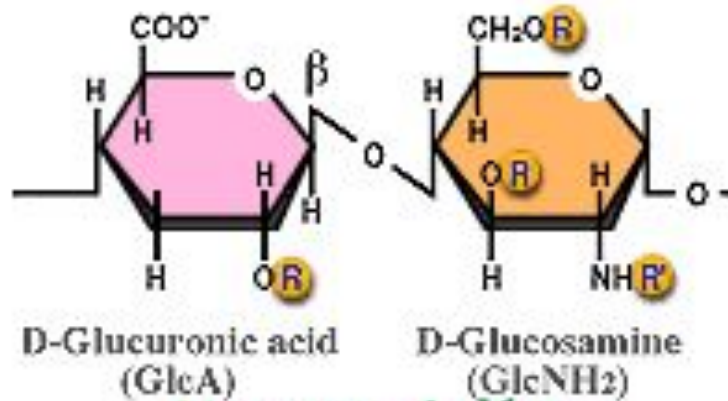
- [URONIC ACID + AMINO SUGAR]_n
- sulfate groups may be present also



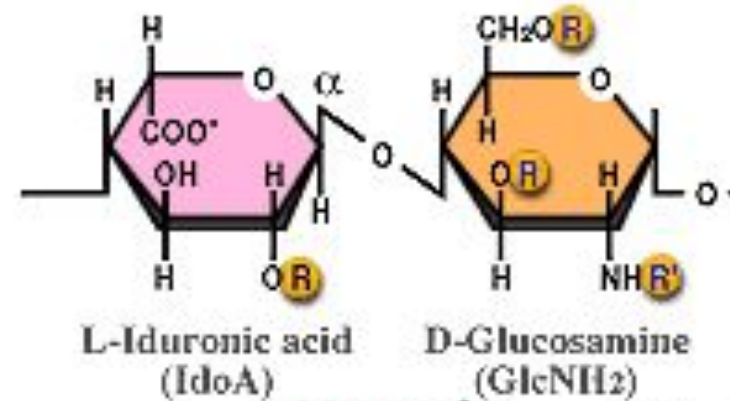
Chondroitin Sulfate



Dermatan Sulfate



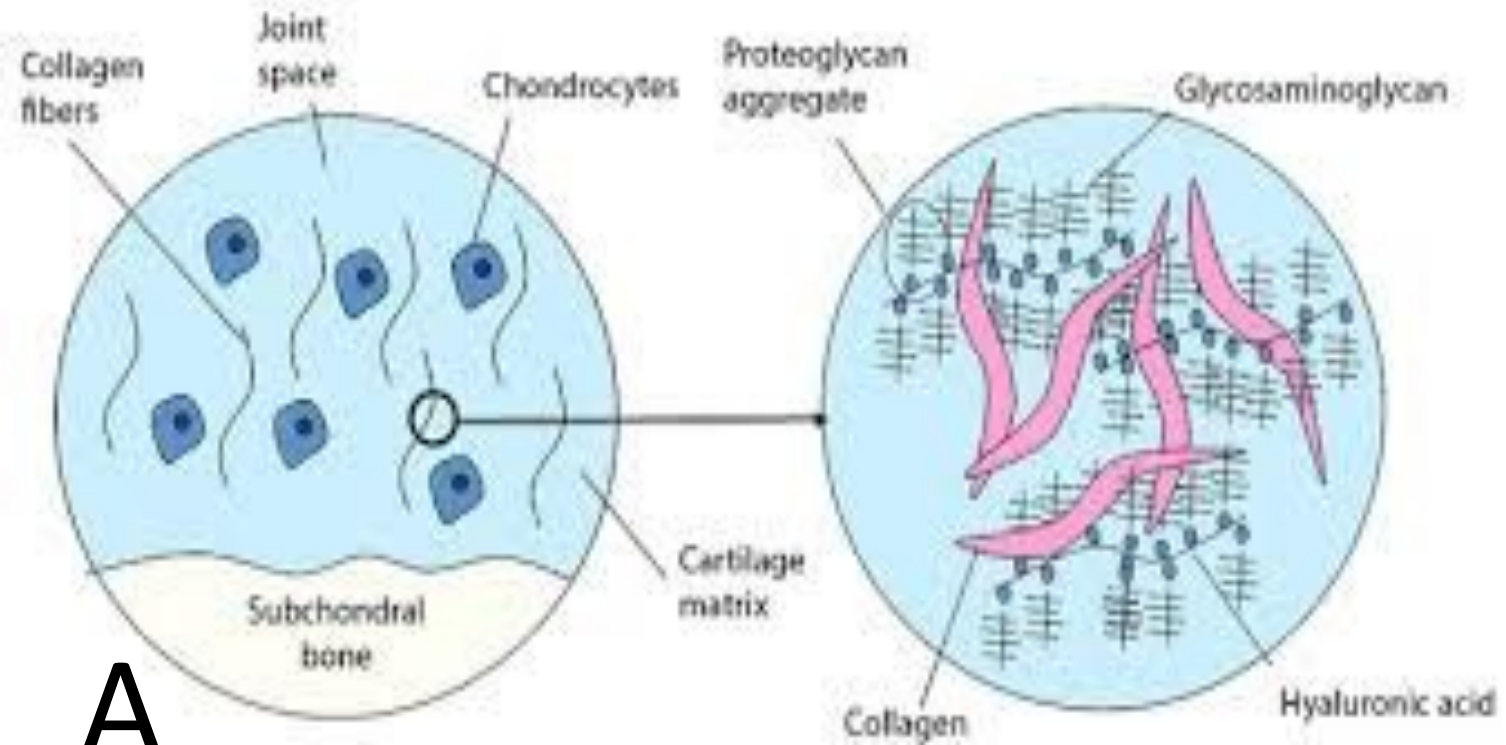
Heparan Sulfate



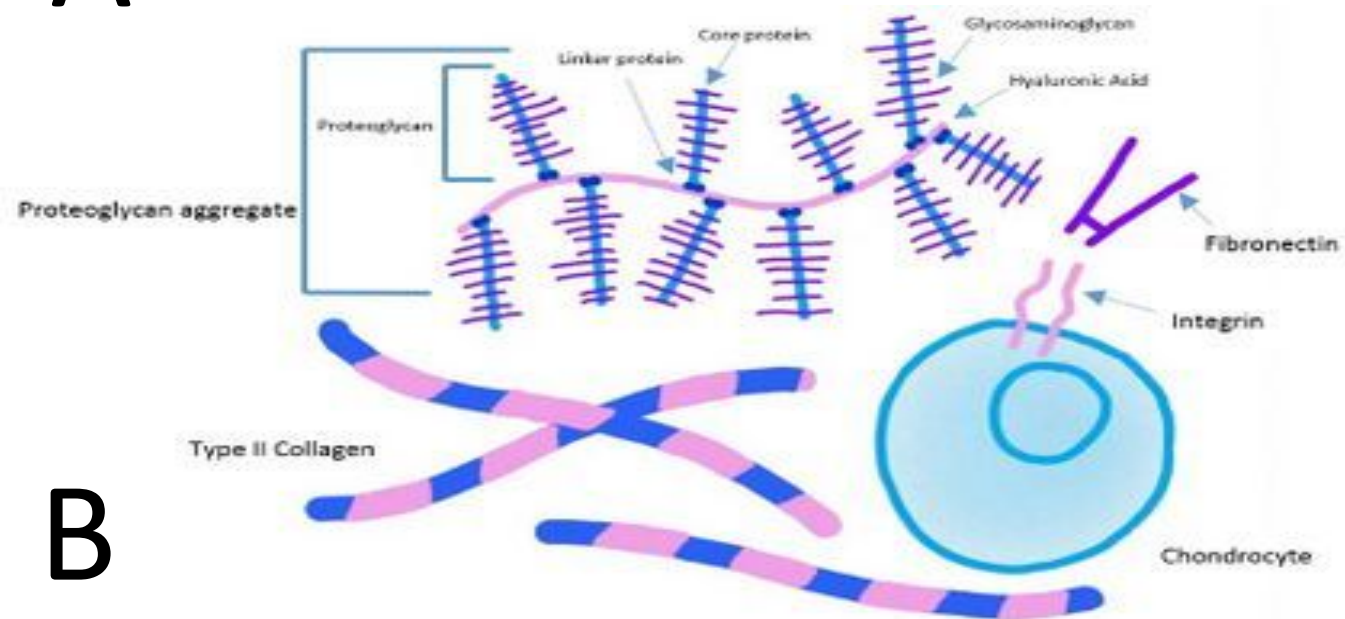
Heparin

cont..

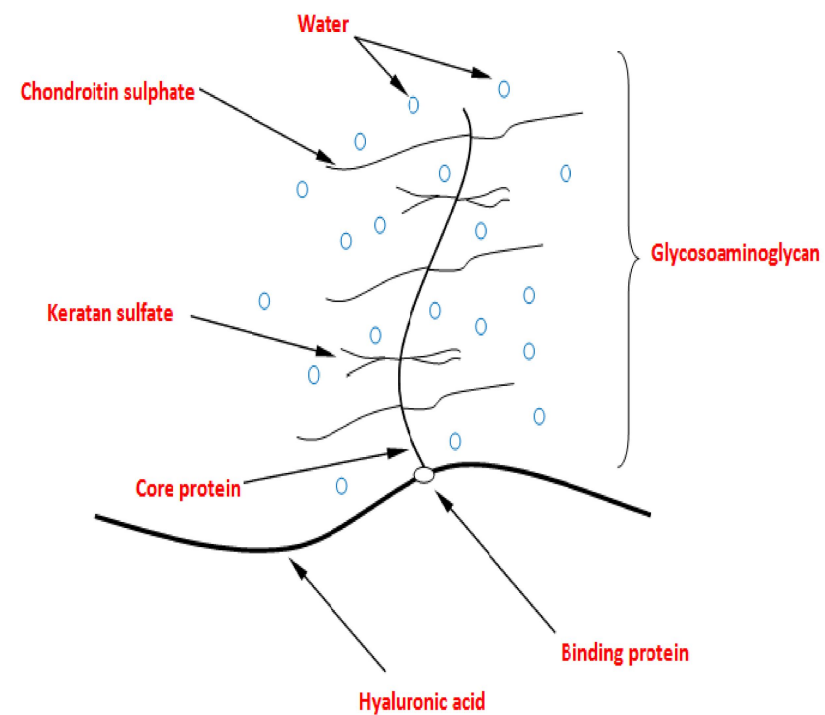
- ✓ Major components of extracellular matrix of connective tissue, including bone and cartilage, synovial fluid, vitreous humor and secretions of mucus producing cells.
- ✓ Gel forming component of extracellular matrix.
- ✓ The anionic groups (carboxy & sulfate groups) being strongly hydrophilic tend to bind large amount of water producing gel like matrix, that forms the ground substance.
- ✓ Due to their high negative charge density, they have a strong tendency to attract **water molecules** and positive ions into the extracellular matrix.



A



B



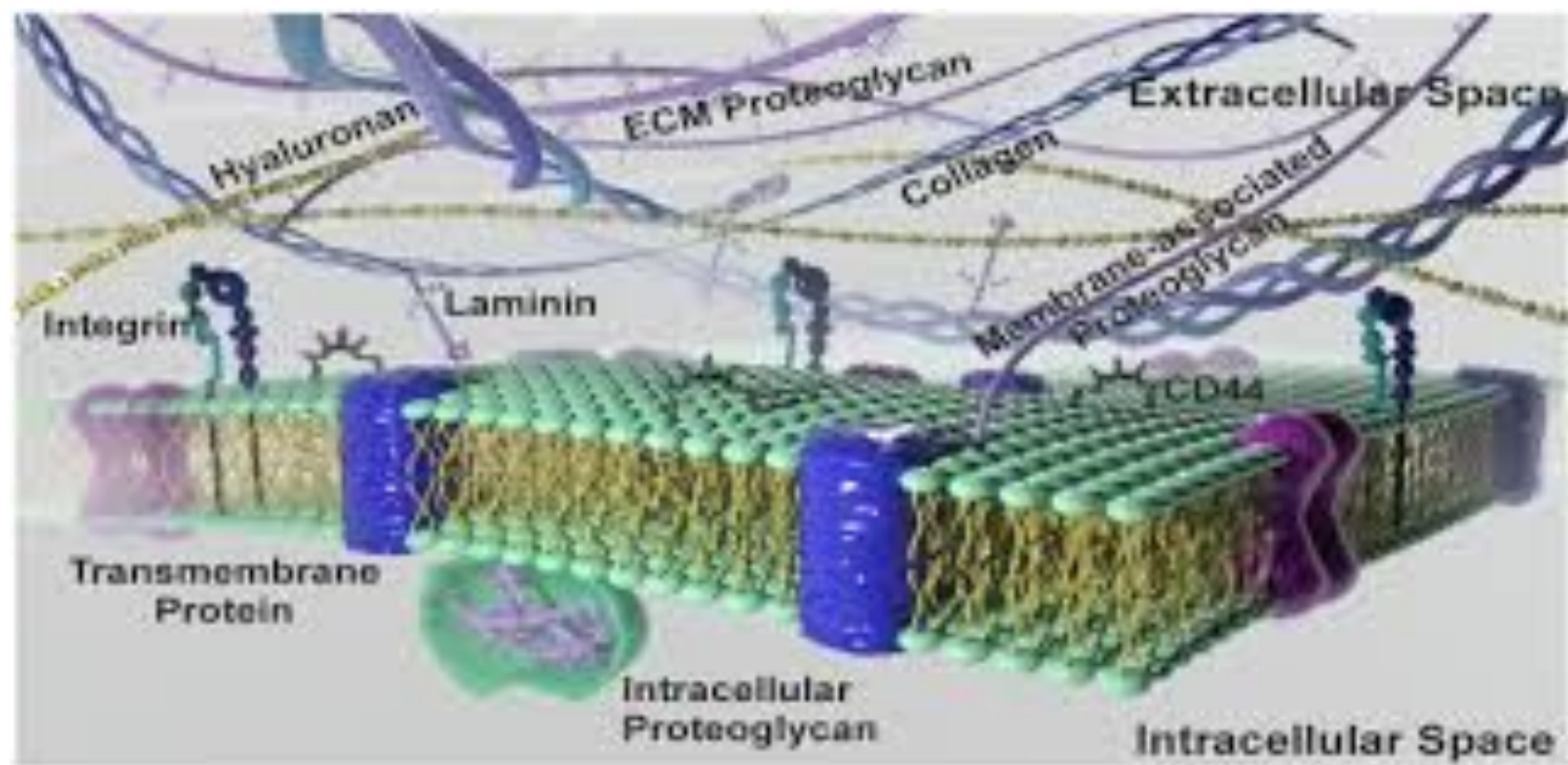
C

FUNCTIONS

- ✓ Structural support to connective tissue
- ✓ GAGs form matrix or ground substance that stabilizes and supports the cellular and fibrous components of tissues.
- ✓ Heteropolysaccharide chains repel one another and therefore exist in extended conformation in solutions.
- ✓ This produces slippery consistency of mucus secretions and Synovial fluid.

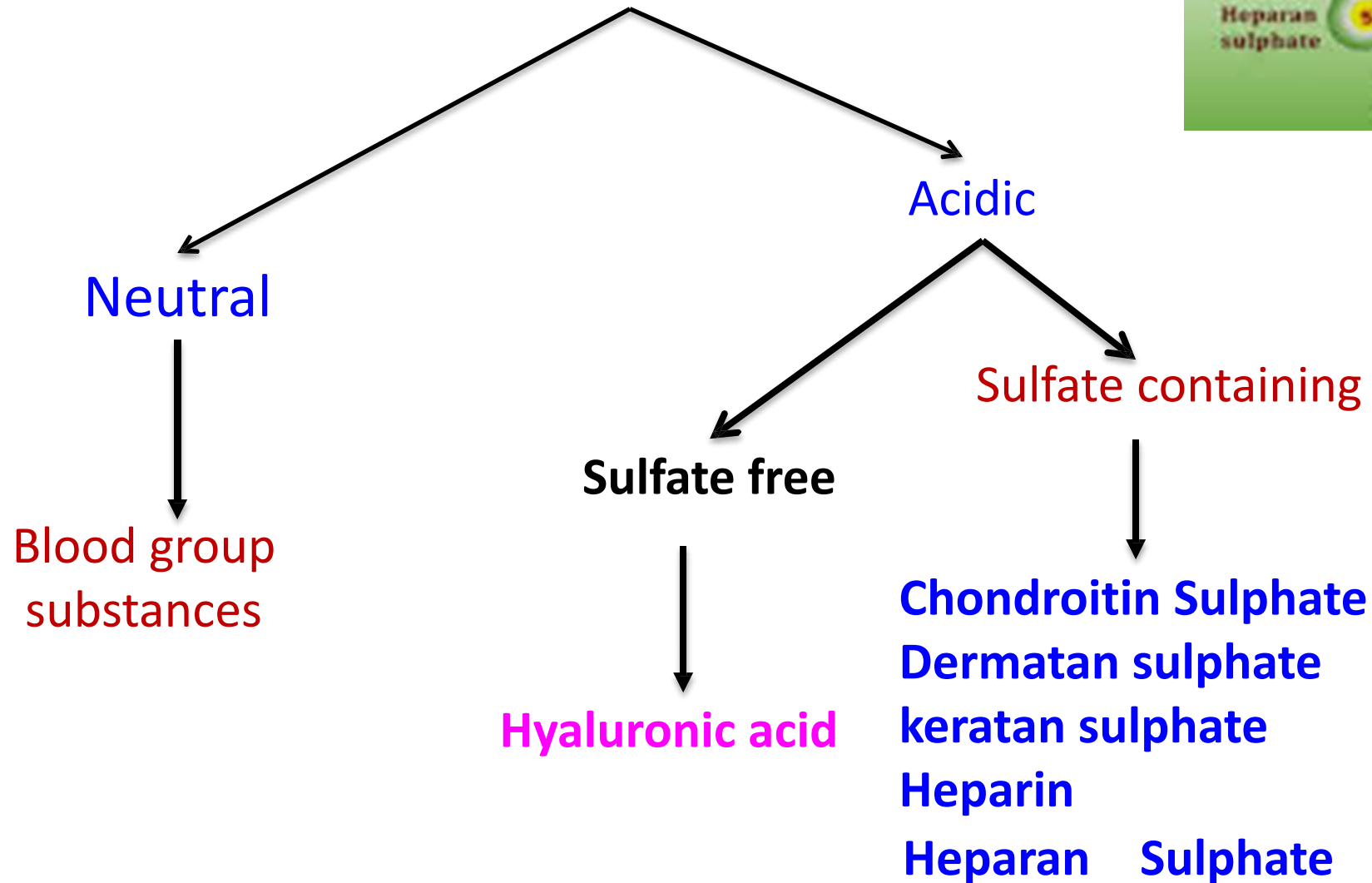
Other functions:

- ✓ Plays an important role in mediating cell-cell interactions
- ✓ Their slippery consistency makes them suitable for a lubricant action in joints.



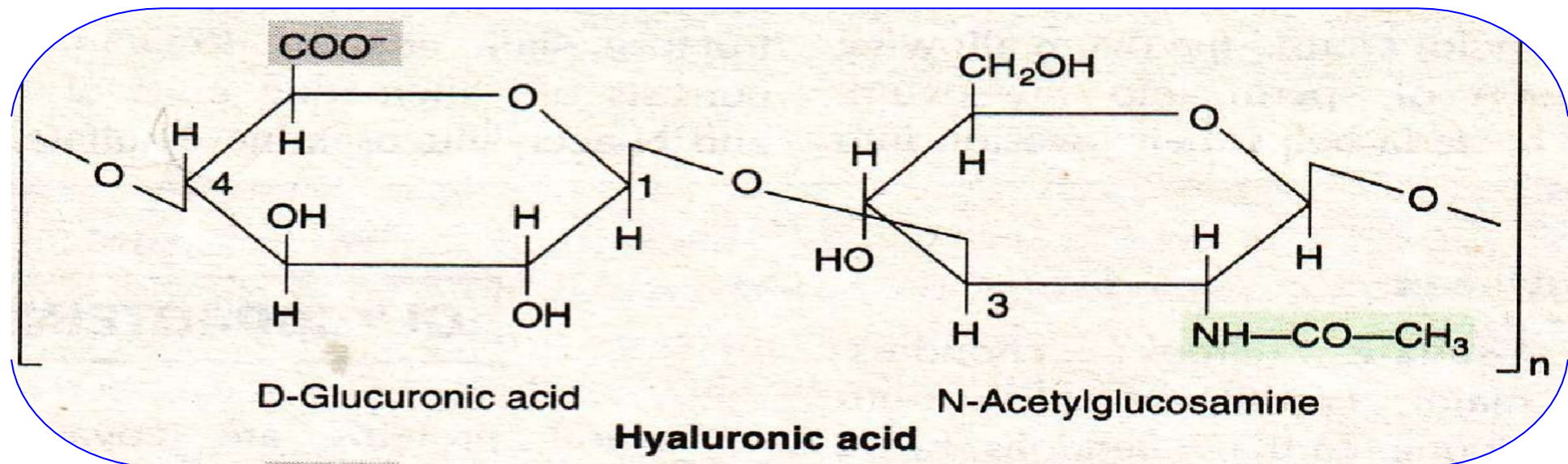
Classification

GAGS



Hyaluronic acid

- ✓ It is sulfate free GAG.
- ✓ Synovial fluid of joints, vitreous humor, connective tissues, skin, umbilical cord, haemolytic streptococci and cartilage.
- ✓ **Composition:** It is composed of repeating units of D-Glucuronic acid and N-acetyl glucosamine.

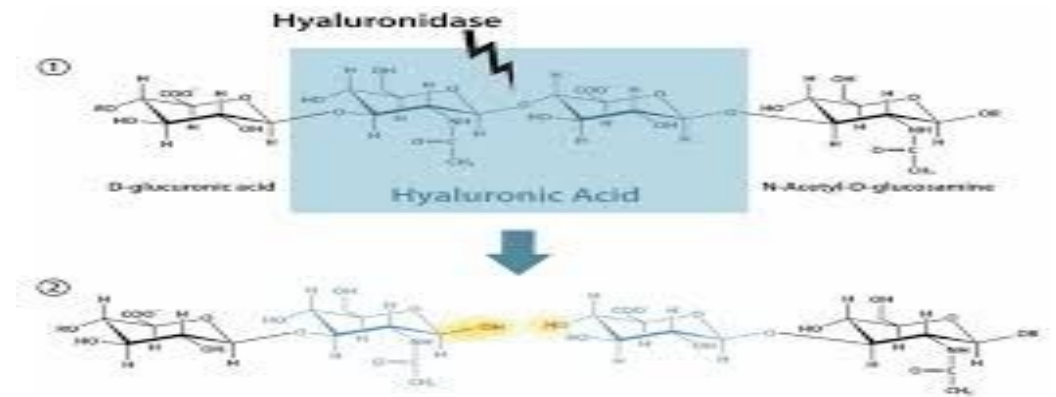


Functions of Hyaluronic acid

- ✓ Serves as a lubricant and shock absorbant in joints.
- ✓ Acts as seives in extracellular matrix.
- ✓ Hyaluronic acid in tissues acts as a cementing substance and contributes to tissue barrier which permit metabolites to pass through but resist penetration by bacteria and other infective agents.
- ✓ Permits cell migration during morphogenesis & wound repair.
- ✓ Hyaluronic acid are present in high concentration in cartilages and have a role in compressibility of cartilage in weight bearing.

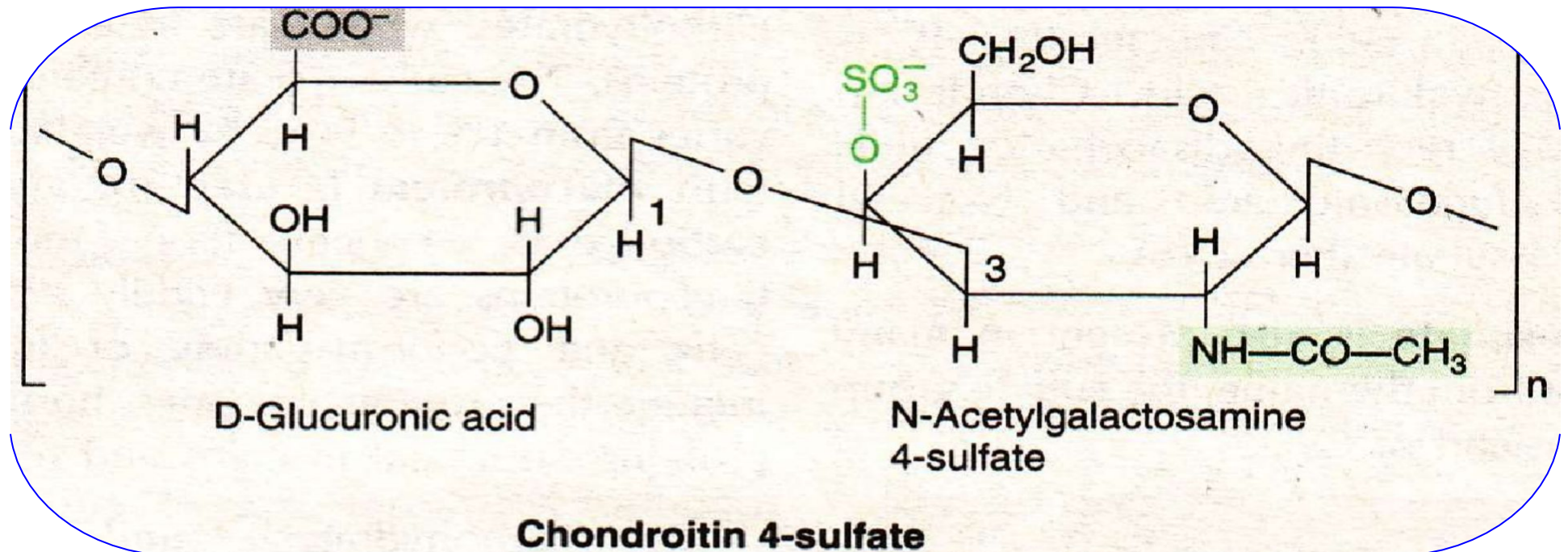
Hyaluronidase:

- ✓ It breaks down the bonds between hyaluronic acid molecules, effectively reducing the "jelly-like" thickness of connective tissue
- ✓ An enzyme that breaks $\beta 1 - 4$ linkages of hyaluronic acid.
- ✓ Present in high concentration in the testicular secretions, it may dissolve the viscid substances surrounding the ova to permit penetration of spermatozoa.
- ✓ The invasive power of some pathogenic organisms may be increased because they secrete hyaluronidase.



Chondroitin 4-sulfate

- ✓ Most abundant GAG in body.
- ✓ Composed of D-glucuronic acid (GlcA) and N-acetyl-galactosamine (GalN).
- ✓ Sulfate groups are added to the GalN units



- ✓ Widely distributed in bone, cartilage & tendons.

FUNCTIONS:

- ✓ In cartilage, it binds collagen & hold fibers in a tight strong network.
- ✓ Role in Compressibility of cartilage in weight bearing.

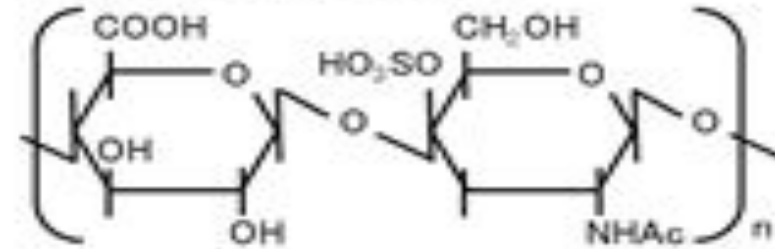
TYPES:

- ✓ **Chondroitin Sulfate A (CS-A):** Also known as **chondroitin-4-sulfate**, major component of hyaline cartilage.
- ✓ **Chondroitin Sulfate B (CS-B):** also referred to as **dermatan sulfate**. it contains L-iduronic acid instead of -glucuronic acid (GlcA): commonly found in skin, tendons, blood vessels, and valves.

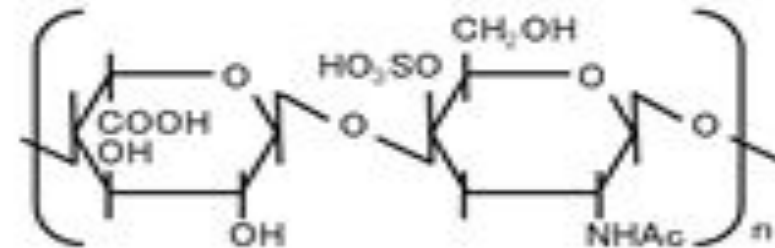
Cont..

- ✓ **Chondroitin Sulfate C (CS-C):**
Known as chondroitin-6-sulfate.
Structure is same as that A except that the SO₄ group is at position 6.
It is found in cartilage and ten-dons.
- ✓ **Chondroitin Sulfate D (CS-D):** Also known as **chondroitin-2,6-sulfate**.
It has been isolated from the cartilage of shark.

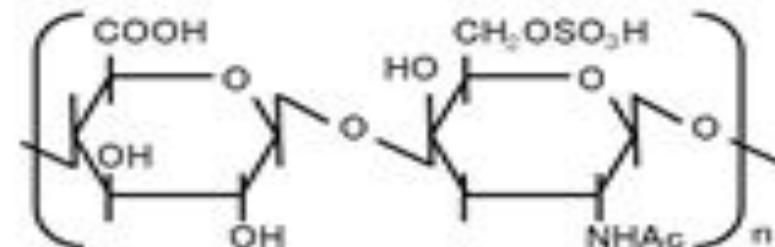
Chondroitin sulfate A
(Chondroitin-4-sulfate)



Chondroitin sulfate B
(Dermatan sulfate)

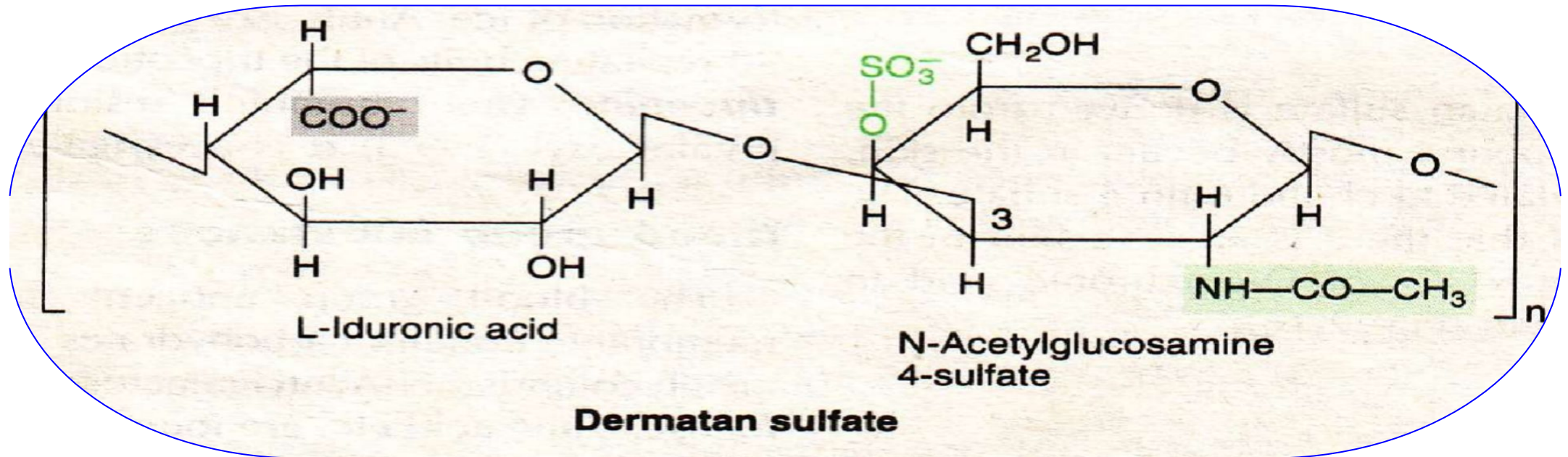


Chondroitin sulfate C
(Chondroitin-6-sulfate)



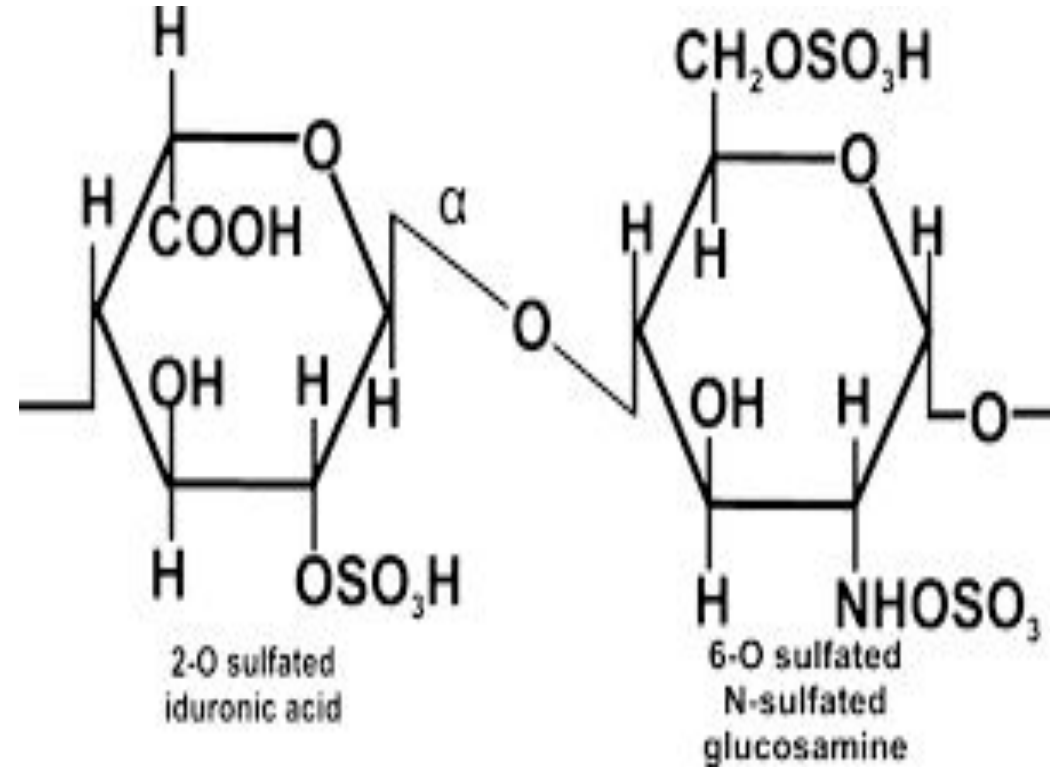
Dermatan sulfate

- ✓ Contains repeating units of L-iduronic acid and N- acetyl glucosamine 4 sulfate.
- ✓ It has a weak anticoagulant property, hence sometimes it is called as β -Heparin.
- ✓ Present in skin, cardiac valves & tendon.
- ✓ Function: Present in sclera of eye where it has important function in maintaining overall shape of eye.



Heparin

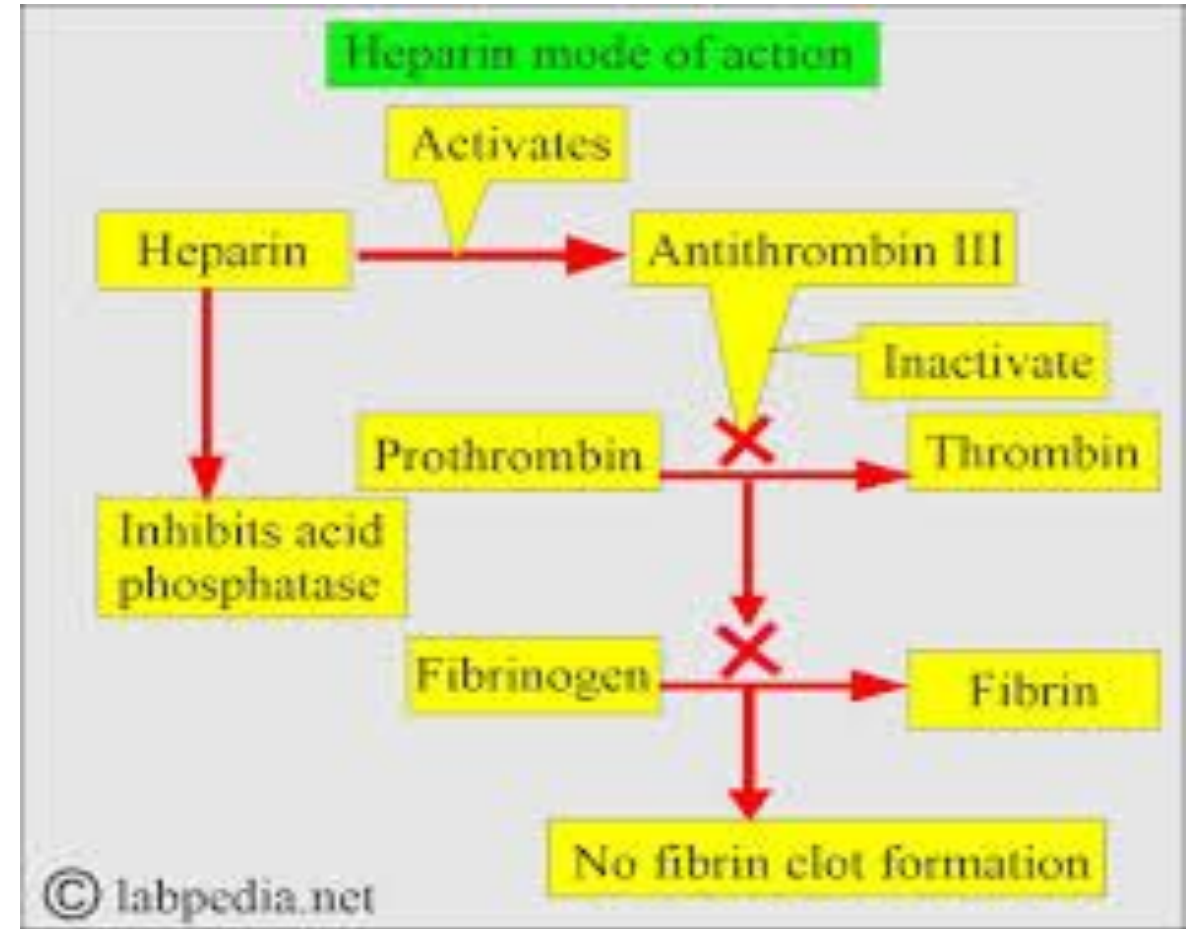
- ✓ Only GAG present intracellular.
- ✓ In granules of mast cells and also in lung, liver and skin.
- ✓ It is also called α -Heparin.
- ✓ It is a polymer of repeating disaccharide units of D-Glucosamine (Glc N) and either of the two uronic acids-D-Glucuronic acid (Glc UA) and L-Iduronic acid (IDUA).



- ✓ Strongly acidic due to presence of more sulfate group.

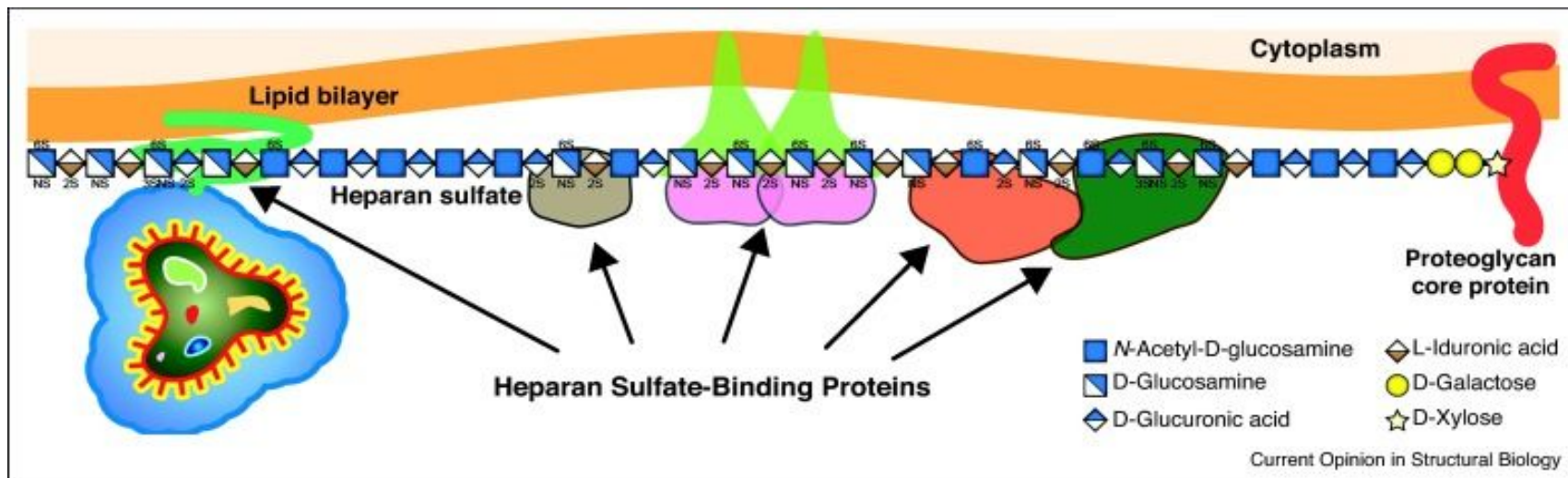
Functions:

- ✓ It is an anticoagulant (prevents blood clotting)
- ✓ Heparin helps in the release of the enzyme lipoprotein lipase (LPL) which helps to clear the lipidemia after fatty meal – so called clearing factor.



Heparan sulfate

- ✓ Structurally similar to heparin, but has a low molecular weight.
- ✓ Contains higher acetyl groups & less sulfate group.
- ✓ Predominant uronic acid is D-Glucuronic acid.
- ✓ It is an extracellular GAG found in basement membrane and is an essential component of cell surfaces.
- ✓ Determines charge selectiveness of renal glomerulus.

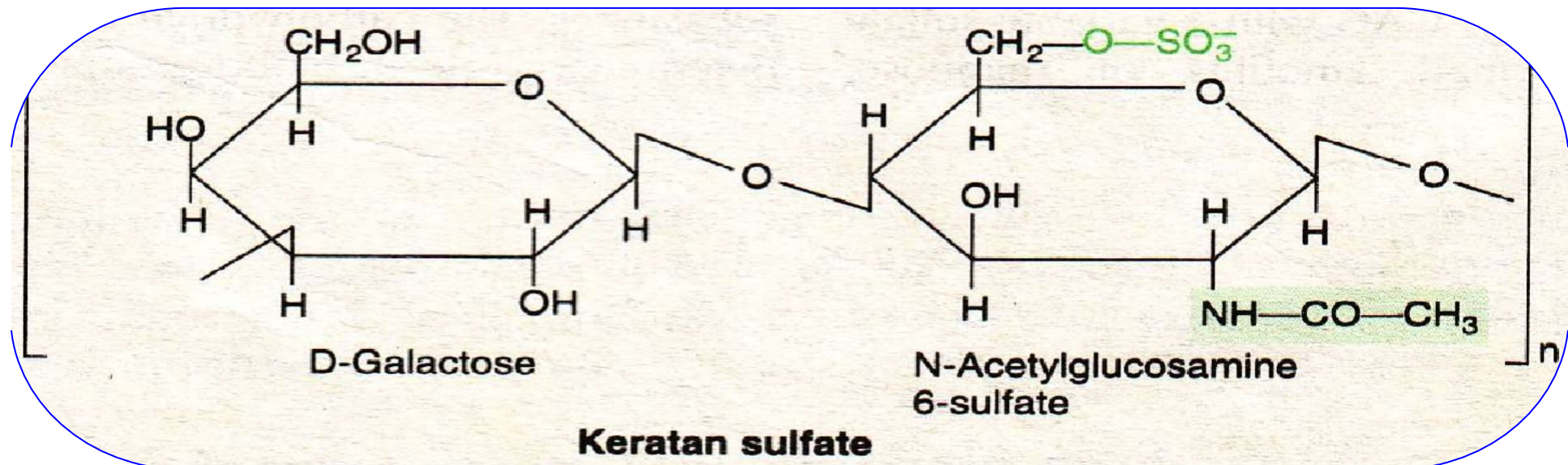


Keratan sulfate

- ✓ Only GAG with no uronic acid.
- ✓ Found in cornea & tendons.

Function:

- ✓ it maintains tissue hydration, provides compressive strength, and supports corneal transparency.

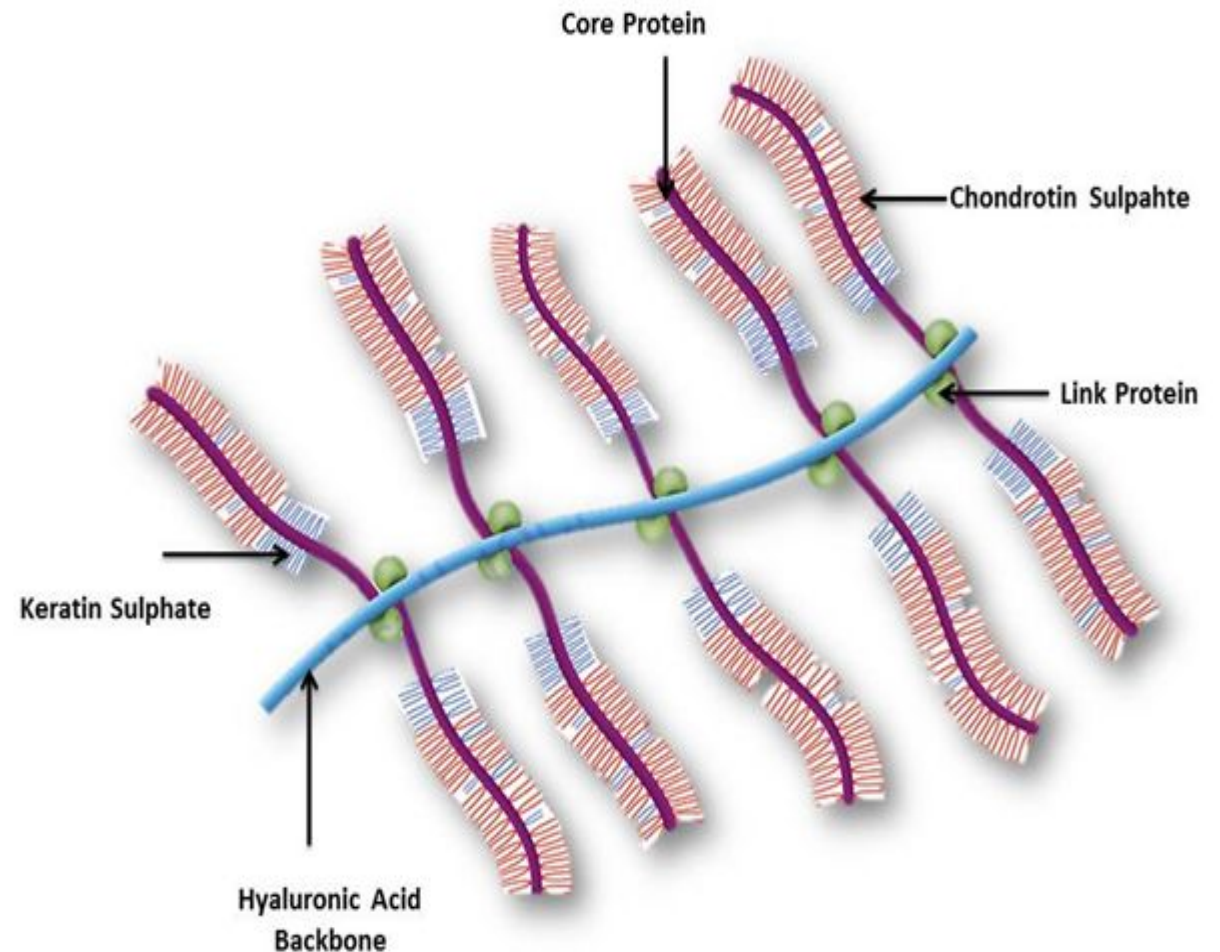


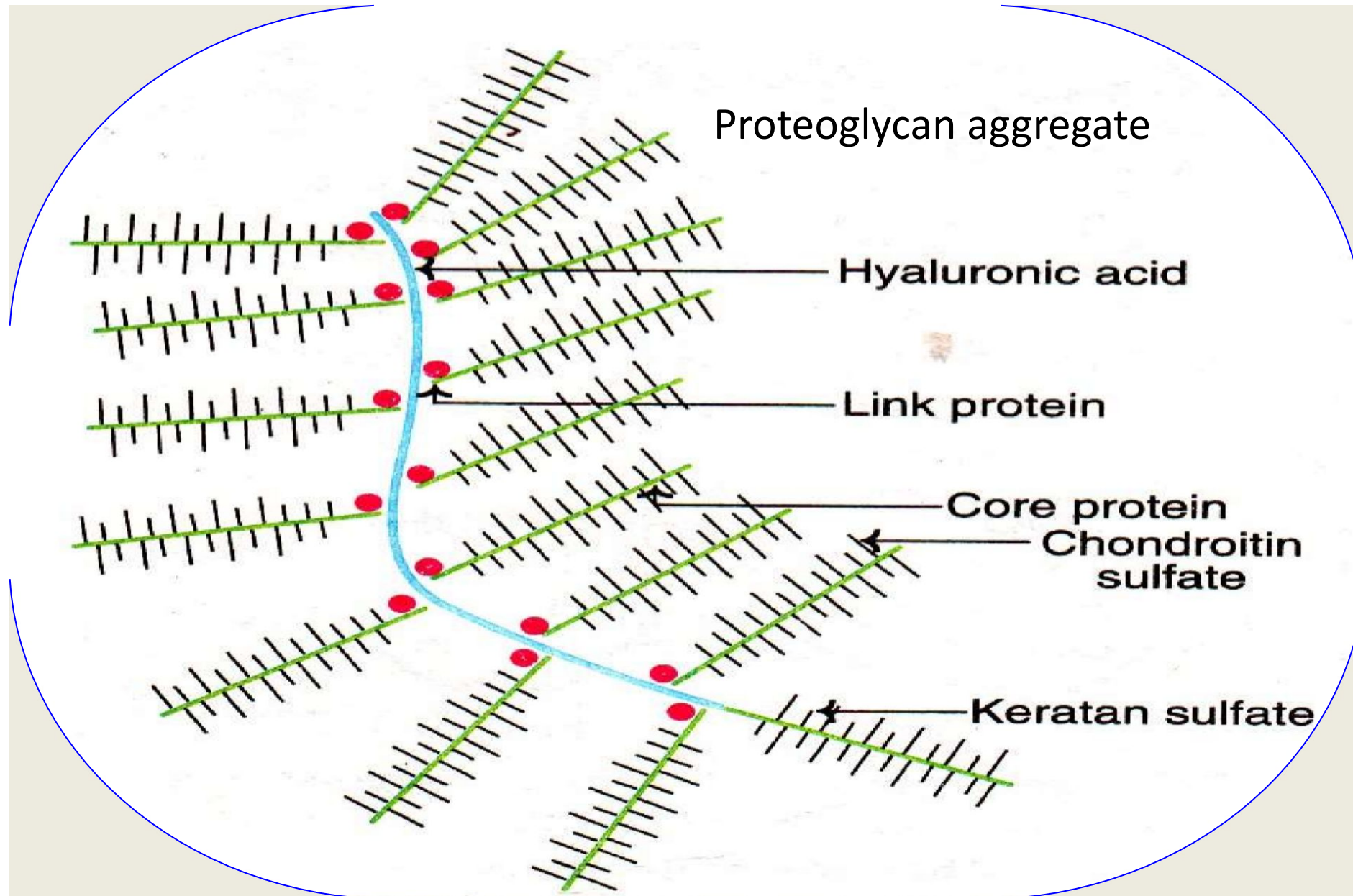
Glycosaminoglycans (GAGs) and Proteoglycans

- Most of the GAGs are covalently conjugated to a protein core, the product of which is termed proteoglycans.
- They are formed mainly of carbohydrates (95%) and only (5%) proteins.
- GAGs are present mainly in the extracellular matrix (ECM) or ground substance in association with other extracellular proteins.

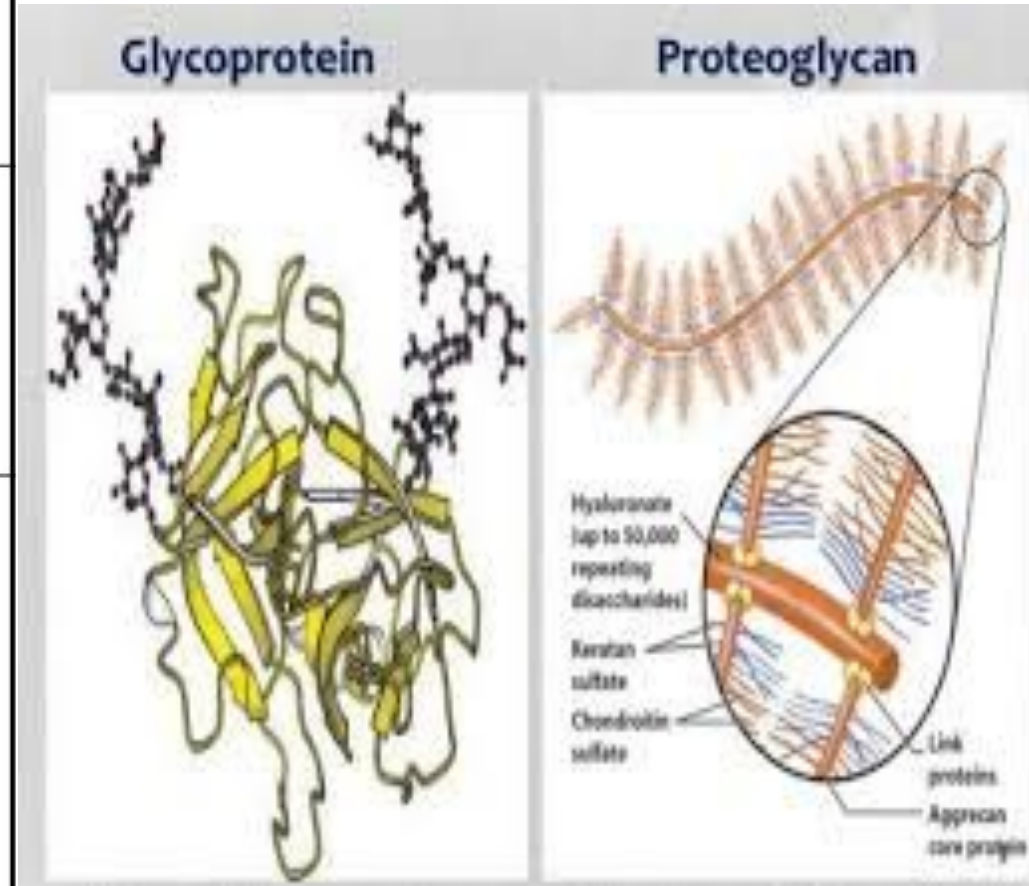
Proteoglycans

- ✓ Proteoglycans are of a class of glycoproteins of high molecular weight that are found especially in the extracellular matrix of connective tissue.
- ✓ Proteoglycans make up a major part of the extracellular matrix, filling the spaces that occur between cells.





Glycoprotein	Proteoglycan
Primarily protein (Protein + some CHO)	Primarily carbohydrate (CHO + some protein)
CHO chains may be negatively charged	CHO chains always negatively charged
Short chains Branched Neutral or -ve No repeating disaccharide units (Oligosaccharides)	Long chains Linear Negatively charged repeating disaccharide units (Glycosaminoglycans, GAG)

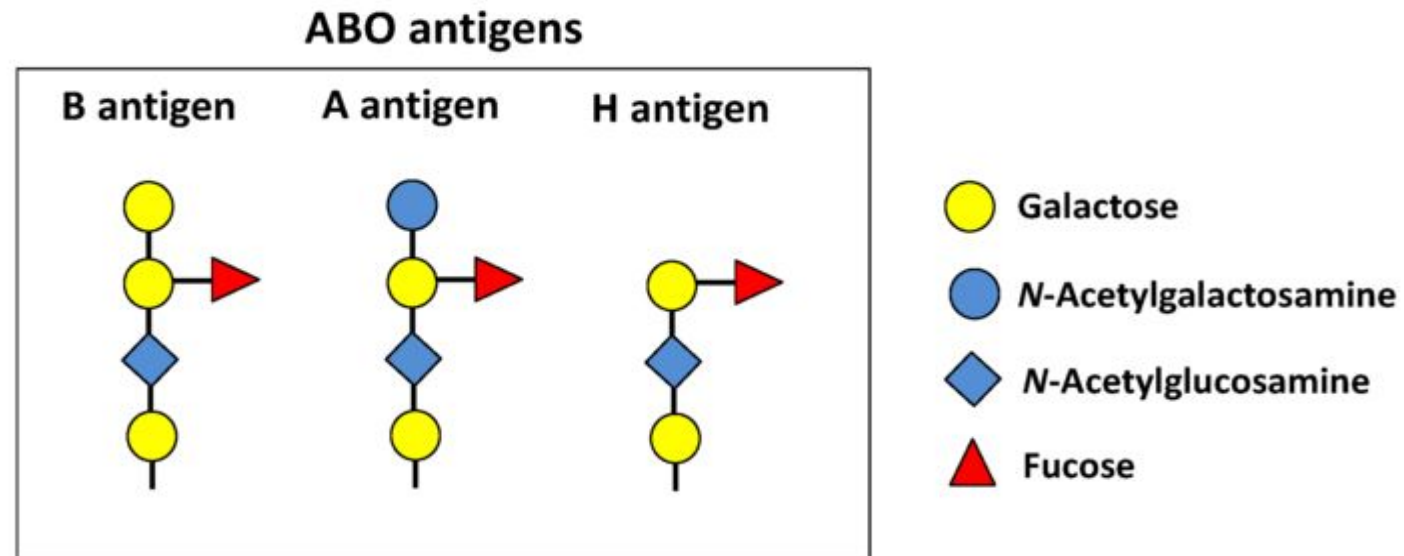


Functions of GAGs and proteoglycans

- ✓ They are important constituents of extracellular matrix.
- ✓ Provides flexible mechanical support for the ECM.
- ✓ Acts as a lubricant in synovial fluid.
- ✓ GAGs solutions act as shock absorbers
- ✓ The proteoglycans interact with a variety of proteins in the matrix, such as collagen and elastin, and these interactions are important in determining the structural organization of the matrix.
- ✓ Hyaluronic acid proteoglycans involved in wound healing (repair). In the initial phase of wound healing, hyaluronic acid concentration increases many folds at the wound site.

Blood group substances (blood gr Antigens)

- RBC membrane contains several antigenic substance, based on which classified into different blood groups.
- They contain carbohydrates as glycoproteins or glycolipids.
- Galactose, fucose, Galactosamine (acetylated) and acetylated glucosamine are found in blood gr substances.
- Carbohydrate content plays a determinant role in blood grouping.



Agar and Agarose

- ✓ Agar is a hydrophilic polysaccharide complex, obtained from sea weeds.
- ✓ It is dissolved in water at 100 °C, which upon cooling sets into a gel.
- ✓ Agar cannot be digested by bacteria so it is widely used as a supporting media to culture bacterial colonies.
- ✓ Also used as a supporting agent for immuno-diffusion and immuno-electrophoresis.
- ✓ Two components :
 - ✓ Agarose (unbranched) 70%
 - ✓ Agaropectin (branched) 30%
- ✓ Agarose is made up of D- galactose combined with 3,6- anhydro L-galactose units and is used as a matrix for electrophoresis.

AGAR PLATES



Thank you